FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN BACOOR CITY

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FloodSense: An Android-Based Expert System for Flood Susceptibility Assessment and Pre-Event Preparedness in Bacoor City

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INTRODUCTION

The first decisions about a flood are often made before floodwater is visible. A household may protect belongings, change a planned trip, or make arrangements for someone who may need help leaving home. These choices depend on how people understand weather information and the places where they live and move. Preparedness thus begins before an emergency response, when there is still time to consider protective action.

Flood conditions are not uniform across a city. Rainfall interacts with elevation, drainage, land cover, and river systems, so a broad weather update cannot describe the same conditions at every location. A Philippine GIS study used several geographic factors to map flash-flood susceptibility within one barangay, illustrating the importance of local context (Puno et al., 2022). A susceptibility map, a rainfall bulletin, and an official warning also communicate different kinds of information. Their distinctions matter when people interpret what they see before a possible flood.

In the Philippines, flood-related information carries different levels of public authority. The Philippine Atmospheric, Geophysical and Astronomical Services Administration issues official hydro-meteorological information, while national and local disaster-management bodies coordinate action under established response

arrangements (National Disaster Risk Reduction and Management Council [NDRRMC], 2024). Information used to consider possible conditions before an event is different from an official forecast, warning, or evacuation instruction. Keeping these roles distinct helps readers understand the purpose and authority of each message.

Bacoor City offers a concrete local setting for examining preparedness. The Japan International Cooperation Agency (JICA, 2021) reported completion of the Imus River Basin subproject, including retarding basins in Bacoor and Imus intended to reduce peak floodwater volume in low-lying Cavite areas. These structures show that flood management in the locality involves river behavior, built infrastructure, and cooperation among public institutions. They also place residents' preparations within a wider setting of public works and local disaster management.

The proposed system brings these distinctions into a single pre-event information flow. A resident confirms a hypothetical rainfall scenario and location, requests a deterministic flood-susceptibility assessment, and receives a human-readable result with separate preparedness guidance. A map places the assessment in geographic context, while permission-controlled administration and source records support responsible handling of the information. These functions are intended to aid preparation and interpretation, not to replace official forecasts, warnings, or evacuation decisions.

Statement of the Problem

Information from public advisories, barangay announcements, online sources, and personal contacts has different purposes and geographic detail. In the initial purposive survey of 40 qualified Bacoor City adults, 29 respondents (72.5%) reported difficulty obtaining needed flood information, and 30 (75.0%) reported difficulty identifying their location's flood susceptibility or did not know how to do so. These findings describe the surveyed group, not all city residents*. What factors limit access to understandable, location-specific flood-susceptibility information before heavy*

rainfall (see Appendix Figure 1. Fishbone Diagram of Location-Specific Flood Information Needs)?

A location-specific assessment requires geographic facts, rainfall-scenario inputs, expert knowledge, and a clear explanation of how a result was reached. Institutional interviews and document review identified source, permission, and validation questions that must be resolved before local information can be treated as authoritative. They did not certify an operational Bacoor susceptibility dataset. *What knowledge, assessment, and governance gaps must be addressed to connect an explainable susceptibility assessment with resident information needs* (see Appendix Figure 2. Fishbone Diagram)?

Preparedness advice and evacuation-center references serve different decisions from a susceptibility classification. Among the 40 surveyed residents, 33 (82.5%) reported difficulty finding verified evacuation-center information or did not know where to find it, while the needs assessment also recorded demand for more situation-specific guidance (see Appendix Table 2. Initial Needs-Assessment Summary). *What limits access to relevant pre-event preparedness guidance and verified evacuation-center reference information for the participating residents* (see Appendix Figure 3. Fishbone Diagram)? The diagrams organize observed needs and factors for investigation; their branches are not independently proven causes.

Objectives of the Study

Generally, the study seeks to develop an application as a scenario-based Android system for explainable flood-susceptibility assessment and separate pre-event preparedness support in Bacoor City.

Specifically, it aimed to:

1. identify the flood experiences, information-access difficulties, preparedness needs, and technology conditions reported by eligible Bacoor City respondents;

2. analyze the geographic, knowledge, rainfall-scenario, privacy, provenance, preparedness, and administrative requirements supported by resident, interview, and documentary evidence;
3. develop an application as a Flutter-based Android resident application and Django-based Web Administration System through Evolutionary Prototyping, with these major functions:
    - authenticated resident and authorized-administrator access with role-based permissions and privacy controls;
    - user-confirmed rainfall scenario and location inputs, including barangay selection, manual map pin, and optional foreground location;
    - deterministic susceptibility assessment, explanation, and geographic presentation;
    - separate preparedness guidance and evacuation-center reference information; and
    - controlled administration of geographic references, parameters, guidance, sources, and review records;
4. test the resident and administrative functions through applicable unit, integration, API, spatial, and end-to-end procedures; and
5. evaluate selected product-quality characteristics of the application using an approved ISO/IEC 25010-based instrument after intended users complete defined system-use tasks.

Theoretical Framework

The theoretical framework of FloodSense connects authenticated access, user-confirmed scenario and location inputs, deterministic susceptibility assessment, geographic presentation of the result, separate preparedness support, and governed administrative information. These functions are organized into six function-named

modules. Database and software technologies support the modules but are not separate conceptual modules.

User Access and Authentication Module. This module controls resident and administrator identity, registration, sign-in, verification, password recovery, privacy and legal acceptance, profiles, roles, and permissions. Authentication is necessary because the application processes account information, optional location data, and governed administrative records that must not be available to unauthorized users.

Scenario and Location Input Module. This module receives the resident's hypothetical rainfall intensity and duration and the selected location. The resident may choose a barangay, adjust a temporary map pin, or request one-time foreground location access, then confirm or correct the resolved barangay before requesting assessment. These inputs describe a user-selected scenario rather than a live observation or forecast.

Susceptibility Assessment Module. This module validates the confirmed scenario and location facts, applies deterministic rule matching and conflict handling, and produces a susceptibility class with its rule basis. The same confirmed facts and rule version produce the same result. Scientific and local validity still depends on appropriate data, provenance, expert review, and institutional authorization.

Map and Results Module. This module presents the susceptibility class, a human-readable explanation, and map-based geographic context for the selected area. It distinguishes administrative reference boundaries from susceptibility information and displays the assessment outcome without changing the Expert System conclusion.

Preparedness Decision Support Module. This module presents source-backed preparedness guidance through structured questions and information flows. It also displays evacuation-center reference records and a straight-line nearest-center ranking when suitable records are available. The guidance remains separate from the

susceptibility assessment and does not modify the assessment class or issue an evacuation order.

Administration and Data Governance Module. This module supports authenticated and permission-controlled management of rainfall and map references, assessment parameters, DSS content, evacuation-center records, data sources, review records, and audit history. Ordinary administrators do not directly modify raw rule conditions, priorities, conflict-resolution logic, or algorithm code. Technical Django Admin access remains separate and is reserved for developer or research maintenance.

System Architecture

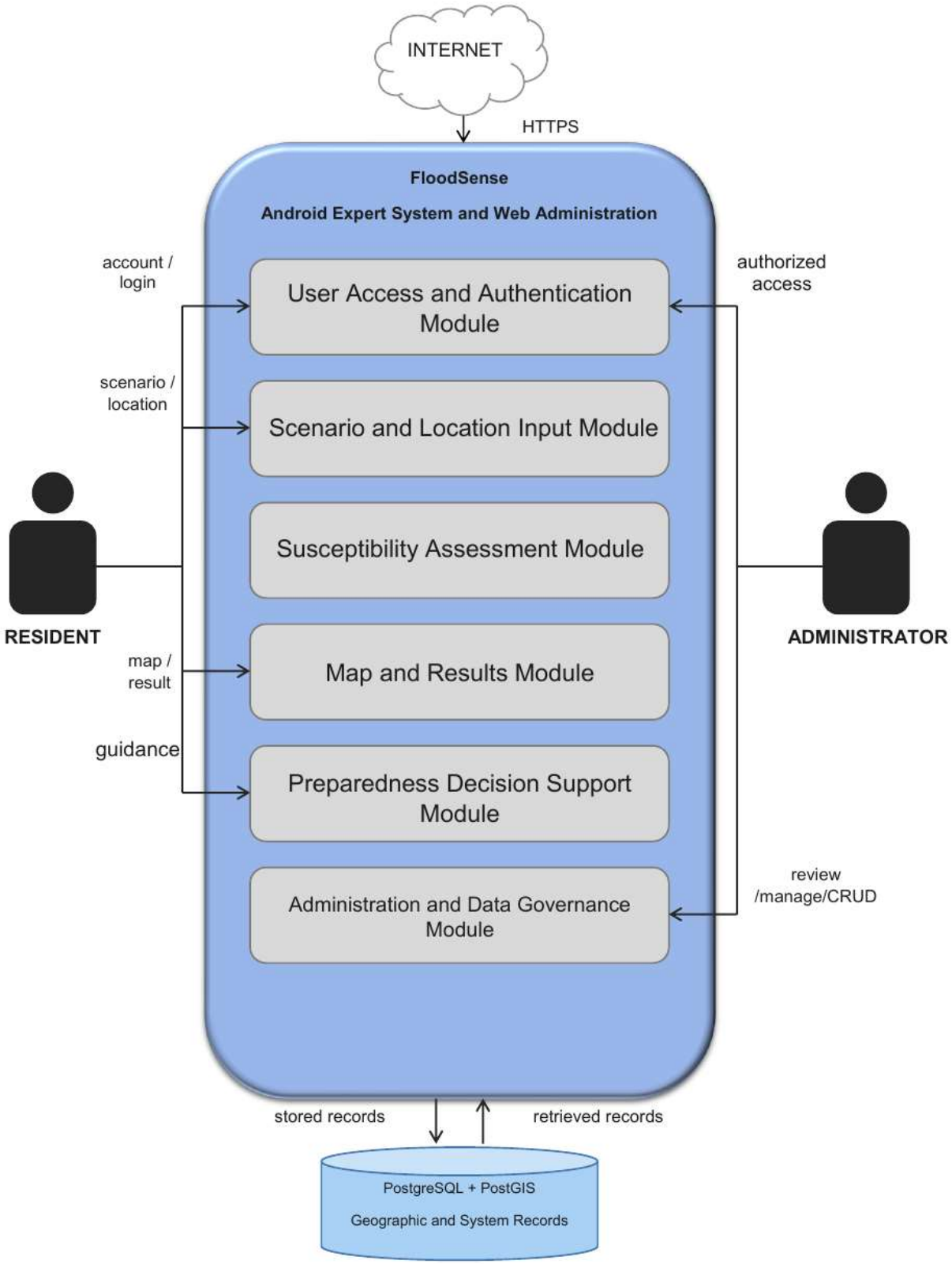


Figure 1. System Architecture of FloodSense.

Significance of the Study

FloodSense contributes an evidence-based approach to scenario-oriented flood preparedness by combining an explained susceptibility assessment with separate guidance and visible information boundaries.

For the **Bacoor Disaster Risk Reduction and Management Office**. The study may provide a complementary channel for presenting validated preparedness and evacuation-center information while retaining source, review, and audit records. It does not replace official warning, response, or emergency-command functions.

For the **City Planning and Development Coordinator Office**. The study demonstrates how appropriately authorized administrative and geographic information may support location-specific public communication while maintaining provenance and validation requirements.

For **Bacoor City residents**. The resident application provides a user-initiated process for confirming a hypothetical rainfall scenario and location, requesting an explained susceptibility result, viewing geographic context, and obtaining structured preparedness information.

For **future researchers**. The study documents the separation of deterministic assessment from preparedness guidance, the importance of authentication and role control, and the distinction among technical processing, scientific validation, and institutional authorization. Its procedures and findings require renewed validation before use in another locality.

Time and Place of the Study

The study is situated in Bacoor City, Cavite, the geographic setting for the resident needs assessment and institutional inquiry. Cavite State University–Bacoor City Campus serves as the academic base for system development and manuscript preparation. The location of the study corresponds to the residents and local offices

whose information needs and responsibilities concern flood preparedness in Bacoor City.

The institutional interviews were conducted at the respective Bacoor City offices of the Bacoor City Disaster Risk Reduction and Management Office (BDRRMO) on May 12 and May 24, 2026, and the City Planning and Development Coordinator Office (CPDCO) on May 18, 2026. These visits preceded the principal study period and provided local institutional context for the Bacoor-focused research.

The principal study period began in September 2026 and is expected to continue through May 2027. The initial resident needs survey was completed on October 11, 2026, with 40 qualified respondents from 14 Bacoor City barangays. The schedule of study activities for September through December 2026 is presented in the Gantt Chart (see Appendix Table 1. Gantt Chart).

Scope and Limitation of the Study

FloodSense comprises a connected Android application for Bacoor City residents and a browser-based administration system for authorized personnel. The study covers pre-event, scenario-based flood-susceptibility assessment and preparedness support within its documented Bacoor administrative coverage. It is intended to help residents examine a hypothetical rainfall condition for a chosen area, rather than report a flood occurring at that moment.

Residents authenticate and select or confirm a hypothetical rainfall intensity and duration. They may choose a barangay, place or adjust a map pin, or request optional one-time foreground location; they then confirm or correct the resolved barangay and explicitly request an assessment. The application presents the selected scenario, an explained susceptibility class, and geographic context through its resident interface. The corresponding resident and administrator interactions are shown in the Use Case Diagram (see Appendix Figure 4. Use Case Diagram of FloodSense).

The deterministic Rule-Based Expert System performs the susceptibility assessment. A separate Decision Support System presents preparedness guidance and evacuation-center reference information without changing the assessment result. The interactive map provides administrative and geographic context for the selected area, while any susceptibility display remains subject to the validity and provenance of its underlying information.

Authorized administrators review rainfall and map references, parameters, preparedness content, evacuation-center records, data sources, and audits through the web system. They cannot edit raw rules, priorities, conflict-resolution logic, or algorithm code. Location access is user-initiated; geographic records require documented sources and validation before authoritative use. The conceptual information exchanges among residents, administrators, and data sources or expert reviewers are shown separately (see Appendix Figure 5. Context Diagram of FloodSense).

However, the system is limited to scenario-based assessment and preparedness support; it does not continuously monitor rainfall or water level, issue official forecasts or warnings, send autonomous alerts or evacuation orders, track location in the background, determine safe routes, report live evacuation-center occupancy, or use machine-learning prediction. Its 47-barangay layer is an administrative reference, not a City-verified flood-susceptibility dataset. Assessment inputs and guidance require appropriate local expert and institutional validation before authoritative use. Evacuation-center distance is a straight-line reference and does not establish road safety, accessibility, opening status, or capacity. Connected functions require a compatible device and network access. The purposive needs-assessment sample of 40 respondents is not representative of all Bacoor residents.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Android Application. The resident-facing FloodSense software developed with Flutter. It includes authentication, legal acceptance, onboarding, scenario and location confirmation, assessment requests, map/results presentation, structured preparedness guidance, nearest-center information, preferences, and account controls. It does not provide live alerts, assessment history, fully offline operation, or official warnings.

Color-Coded Susceptibility Levels. The visual scheme used on the dynamic map in which green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility for the active scenario.

CPDCO (City Planning and Development Coordinator Office). The Bacoor City office that maintains planning and geographic information, including elevation, land-use, flood-hazard, and barangay-boundary materials relevant to the study.

Decision Support System (DSS). The component separate from the Rule-Based Expert System that presents published preparedness guidance and structured questions using the assessment context. It does not change the susceptibility result, issue an official warning or evacuation order, or make the user's final decision.

Expert System. A knowledge-based computer system that applies stored facts and conditional rules through an inference process. In FloodSense, it produces an explainable susceptibility class for a supported geographic area and rainfall scenario.

Flood Susceptibility. The relative potential of a supported geographic area to experience rainfall-related flooding under a user-confirmed scenario. FloodSense expresses the assessment as Low, Moderate, High, or Very High and identifies it as a research-system output rather than an official Bacoor City classification or real-time prediction.

Geographic Information System (GIS). A system for organizing, analyzing, and presenting information associated with geographic locations. FloodSense uses GIS functions for polygons, points, containment queries, interactive visualization, and evacuation-center locations.

GeoDjango. The geographic extension of Django used by FloodSense to represent spatial data and perform geographic operations between backend services and the PostGIS database.

Inference Process. The procedure that matches facts from the selected rainfall scenario and geographic area against validated rules in the knowledge base to derive a susceptibility conclusion and supporting explanation.

Knowledge Base. The structured collection of documented facts, conditional rules, geographic attributes, rainfall parameters, susceptibility categories, and provenance records used by the Expert System. These materials require appropriate expert and institutional validation before they can be treated as authoritative local evidence.

Rainfall Scenario Input. A user-selected hypothetical rainfall intensity and duration used to request a pre-event assessment. The choices are scenario labels, not automatically updated observations, official forecasts, or official warning thresholds.

PostGIS. The spatial extension of PostgreSQL used to store and query FloodSense geographic polygons, points, boundaries, selected locations, and evacuation-center coordinates.

Preparedness Provenance. Information identifying the source, validation status, responsible actor, relevant date, and version of a preparedness item or related record. Provenance permits users and administrators to distinguish approved, provisional, and unsupported information.

Rainfall Parameters. The supported combination of rainfall intensity and duration selected by the user and applied as scenario facts during Expert System assessment.

Rule-Based Expert System. An Expert System in which domain knowledge is represented as conditional rules. FloodSense evaluates applicable rules against known scenario and geographic facts to produce a susceptibility assessment.

Web Administration System. The custom browser-based interface used by authorized administrators for dashboards and permission-controlled workflows involving map and rainfall references, parameters, DSS content and structured flows, evacuation centers, data sources, review records, and audit history. It does not expose raw rule editing. Technical Django Admin is separate and is reserved for developer or research maintenance.

REVIEW OF RELATED LITERATURE

This chapter reviews foreign and Philippine literature and system studies that inform FloodSense's assessment and preparedness-support design. It considers the credibility of flood information, explainable rules, geographic and location data, and the communication of results.

Related Foreign Literature

Flood Assessment Credibility and Geographic Validation

Merz et al. (2024) argued that flood hazard and risk assessments should be validated for the decisions they are meant to support, rather than judged only by whether the calculations run. They organized validation at process, outcome, and impact levels and emphasized suitable data, transparent assumptions, and stakeholder participation. Thus, a repeatable result and a technically valid geographic boundary are necessary checks but do not alone establish the scientific credibility of a susceptibility class.

Their perspective concerns flood assessments generally, not a resident application, and supplies no Bacoor City parameters. For the proposed assessment, their framework supports documenting source records and geographic checks, seeking appropriate expert review, and stating limitations before a scenario result informs preparedness. In particular, an administrative reference boundary must not be presented as an approved flood-susceptibility dataset (Merz et al., 2024).

Rule-Based Decision Support and Explainable Inference

Credible inputs must also be connected to an understandable inference process. Papadopoulos et al. (2022) systematically reviewed 15 rule-based clinical decision-support publications. They described how encoded domain knowledge, case facts, and rule engines can produce case-specific advice, but found uneven use of standards and limited evidence of evaluated clinical outcomes. Although their field is health care, the review helps distinguish a knowledge base, fact matching, and an explanation of the resulting advice.

For the proposed rule-based assessment, the same governed geographic and rainfall facts should yield the same susceptibility result under the same rule version. An explanation should identify the matched basis and its limits, while ordinary administrators remain unable to rewrite raw conditions, priorities, or conflict handling. The clinical rules, outcomes, and evaluation evidence cannot be transferred to flooding or used to claim assessment accuracy in Bacoor City (Papadopoulos et al., 2022).

Actionable Flood-Risk Communication and Preparedness Support

An explainable result is useful to residents only when its meaning and possible actions are communicated clearly. Mostafiz et al. (2022) examined 50 freely available web-based flood-information portals alongside literature on mass media, citizen science, and social media. They found that many outlets addressed short-term information needs, whereas individualized information for longer-term planning remained limited. Their review separates the display of a map or warning from information that helps a person consider feasible protective actions.

This distinction informs the resident application's presentation of a confirmed hypothetical rainfall scenario, a susceptibility result with human-readable context, and separate preparedness guidance. The review does not provide a universal guidance formula, and its portals differ in jurisdiction, authority, and data from Bacoor City. These findings do not make a scenario-based research assessment an official forecast or autonomous warning service (Mostafiz et al., 2022).

Spatial Data Management with PostGIS

Location-specific information also depends on how geographic records are managed. The PostGIS Development Group (2025) documents geometry and geography types, spatial indexes, coordinate transformations, validity operations, and spatial relationship functions within PostgreSQL. These functions allow boundaries and location-linked records to be stored and queried alongside other application data. They are relevant to preserving geographic identifiers and relating resident-selected locations to administrative reference areas.

Database capability must not be confused with authority over the data. PostGIS is a spatial technology, not a susceptibility method or an approved dataset; storing a polygon and passing a geometry check do not verify its institutional origin or the scientific meaning of an assessment. The proposed system therefore requires separate provenance and authorization controls for boundaries, rainfall references, and assessment parameters (PostGIS Development Group, 2025).

Foreground Location and User Choice

Device location introduces a separate issue from spatial storage: when and how residents permit access to their position. Android Developers (2026) distinguishes foreground from background location access and explains approximate and precise permissions. A request made while a person actively uses an assessment differs from continual tracking. The platform guidance also recognizes that a user may grant only approximate location when the application's function permits it.

In the resident application, foreground location is an optional input to a user-requested assessment, while manual location selection remains available. The permission guidance informs this design choice but does not prove that every device will identify the correct barangay. Device accuracy, boundary quality, resident confirmation, and privacy controls require separate checks (Android Developers, 2026).

Related Local Literature

Philippine Flash-Flood Susceptibility Mapping

Philippine mapping research makes the limits of geographic transfer concrete. Puno et al. (2022) mapped flash-flood susceptibility in Barangay Bukang Liwayway, Kibawe, Bukidnon using GIS and frequency-ratio analysis. Their publication described 160 recorded flood points, nine conditioning factors, remotely sensed land-cover inputs, and field checks with local residents. The resulting classes therefore depended on particular observations, selected factors, processing steps, and validation, rather than on map coloring alone.

That geographically isolated barangay has terrain and data conditions unlike Bacoor City's urban setting. Its frequency-ratio weights, classes, and mapped results cannot be adopted as the proposed system's rules without local evidence and expert validation. The study instead supports documenting provenance and distinguishing a research susceptibility layer from an administrative reference boundary (Puno et al., 2022).

Flood Mitigation in Cavite and the Imus River Basin

Within Cavite, the Japan International Cooperation Agency (2021) reported the inauguration of Imus River Basin flood-mitigation facilities, including retarding basins associated with Bacoor and Imus. Its account explained their intended storage of storm runoff and noted overflow affecting low-lying tributary areas. The report establishes the relevance of watershed behavior and infrastructure to the local setting. It is not a resident-system evaluation or a barangay-level susceptibility dataset.

The existence of those facilities cannot establish a household's exposure or the conditions at every Bacoor location. Nor does the report provide assessment rules or a current, validated local susceptibility layer. It supplies geographic context for the research while showing why engineering mitigation and a public-facing, hypothetical-scenario assessment answer different questions (Japan International Cooperation Agency, 2021).

UP NOAH and Public Hazard Maps

Local access to geographic hazard information is illustrated by UP NOAH. Dabu (2024), writing in a University of the Philippines institutional article, described its online display of flood, landslide, and storm-surge hazards for a user's area of interest. This example shows how place-oriented hazard information can be made accessible to the public. The article is institutional literature, not an IMRaD evaluation of the platform, and supplies neither a software-development model nor Bacoor-specific assessment rules.

UP NOAH is therefore a useful reference for map-based risk communication, but its documented purpose and authority differ from FloodSense. The latter asks a

resident to confirm a hypothetical rainfall scenario and request a deterministic susceptibility assessment with separate preparedness guidance. That research output must not be presented as an official forecast or warning, and the institutional article does not establish that NOAH data were incorporated into the proposed application (Dabu, 2024).

HazardHunterPH and Institutional Data Authority

The authority of a geographic result is clearer when its sources are identified. DOST-PHIVOLCS (2023) described HazardHunterPH as a GeoRisk Philippines application for site-specific multi-hazard assessment using official hazard and related information from participating Philippine agencies. Its presentation illustrates the importance of naming sources and specifying an assessment's intended use. The cited account is official platform documentation, not a peer-reviewed system study with a transferable evaluation method.

The proposed research application can draw on that source transparency and place-based presentation without claiming equivalent institutional status. A site-specific assessment backed by named agencies is not interchangeable with an unapproved project layer. Likewise, technical geometry validation of a 47-barangay administrative reference does not make it an official flood-hazard map (DOST-PHIVOLCS, 2023).

Philippine Guidance on Mobile Permissions

Place-based information also requires attention to residents' control over device permissions. The National Privacy Commission (2021) advised developers to request minimum permissions, ask in context, and preserve user choice when optional access is denied. Although its advisory concerned instant-messaging applications, it explicitly discussed location sharing among app permissions. The principle helps assess whether a resident location request is proportional to a specific task.

For the resident application, this means seeking foreground location only during an active, user-initiated assessment and retaining manual alternatives. The advisory does not demonstrate that the application has passed a privacy audit or that

an individual GPS reading is accurate. Implementation, consent behavior, and location accuracy require their own testing evidence (National Privacy Commission, 2021).

Related Foreign Studies

Design and Implementation of Real-Time Flood Early Warning System (FEWS) Based on IoT Blynk Application: An Arduino- and GSM-Based Mobile Flood-Warning System

Wisudawan (2021) investigated a mobile-connected Flood Early Warning System for a reservoir at Universitas Islam Indonesia in Sleman, Yogyakarta. The work addressed the need to observe local environmental conditions and communicate warning levels to a mobile user. Its purpose was continuous site monitoring, not the assessment of a hypothetical rainfall scenario.

The researcher assembled an Arduino Nano with rainfall, water-level, temperature, and humidity sensors, GSM communication, and the Blynk Android application. The prototype was checked in sensor tests and then observed at the reservoir for 30 days. Readings were categorized into Normal, Advisory, Watch, and Warning states; the source did not name a formal software-development life-cycle model.

During the observation period, the reported mean temperature was 28.6 degrees Celsius, mean humidity was 63.7%, and mean water level was 54.5 centimeters. The paper documented transmission and display of readings and warning states. No flood occurred during field observation, so the study did not demonstrate warning performance during an actual flood event.

The prototype illustrates how environmental measurements can be presented understandably on a mobile interface. Its one-reservoir setting, short observation period, and sensor thresholds limit transferability to other places. FloodSense instead uses a resident-confirmed hypothetical scenario and deterministic susceptibility rules; the Indonesian sensor readings cannot validate Bacoor City conditions.

A User Experience Evaluation of a Mobile Application for Disseminating Site-Specific Impact-Based Flood Warnings: The A4alerts App: A Mobile Application for Site-Specific Impact-Based Flood Warnings

Meléndez-Landaverde and Sempere-Torres (2025) examined A4alerts, a mobile application for disseminating site-specific, impact-based flood warnings in Catalonia, Spain. Their research asked whether users considered the application and its warning-related functions useful in an emergency context. The app drew on an operational warning environment rather than a stand-alone hypothetical susceptibility exercise.

Seventy-six civil-protection, emergency-response, and local stakeholders completed an online interactive scenario experiment between January and March 2022. The researchers assessed perceived usefulness, functional appropriateness, and willingness to adopt the app and compared feature rankings using Friedman's test and Bonferroni-adjusted Wilcoxon comparisons. The participant experience was a demonstration, not observation of behavior during a real flood.

Ninety-seven percent of participants considered A4alerts useful during flood emergencies, while 93% said they would use or recommend it. Site-specific impact warnings, official regional warnings, and a self-protection checklist were among the most valued functions. These are reported perceptions of the recruited participants, not evidence of an observed reduction in flood losses.

The study supports careful presentation of place-specific information and actionable guidance. Its stakeholder sample, Catalan warning infrastructure, and scenario experiment constrain generalization to Bacoor City residents. FloodSense can draw on its communication lessons but does not disseminate official operational warnings or inherit the evaluated app's performance.

Real-Time Flood Forecasting and Warning: A Comprehensive Approach toward HCI-Centric Mobile App Development: A Human-Centered Mobile Flood-Forecasting and Warning Prototype

Alsabhan and Dudin (2023) developed and evaluated a human-computer-interaction framework and prototype for mobile real-time flood forecasting and warning. The study focused on the tasks and information needs of front-end users, weather-

station users, and administrators. Its research problem concerned a multi-role forecasting environment rather than pre-event assessment initiated by a resident.

Activity theory and scenario-based design guided nominal-group sessions with five prospective users and five experts. Participants generated 63 situations, from which 12 priority scenarios informed requirements and prototype tasks. The researchers evaluated task performance and satisfaction across usefulness, ease of use, learnability, satisfaction, and overall experience; no Agile, Waterfall, or other SDLC was stated as the study's development model.

The process produced 87 initial requirements and six additional requirements after testing, for a total of 93. Mean satisfaction-category scores ranged from 3.93 to 4.08, although some analytical tasks were harder for participants. The reported prototype and evaluation therefore give evidence about requirements and interaction quality, not about forecast accuracy in Bacoor City.

This study shows why role-specific access and understandable interactions matter in a flood-information application. Its small participant groups and forecasting architecture limit wider inference. FloodSense shares the need for comprehensible outputs and permissions, but its deterministic, resident-requested susceptibility assessment is separate from real-time forecasting and warning.

Developing a Climate-Smart Web GIS App for Multi-Hazard Early Warning against Climate-Based Disaster Risks: A Tanzanian Web GIS for Flood and Drought Early Warning

Msusa et al. (2025) investigated a climate-smart Web GIS application for multi-hazard early warning in Tanzania. Flood analysis focused on Dar es Salaam, while drought analysis focused on Dodoma. The authors sought to bring modeled hazard information and local knowledge into an accessible geographic application for planning and warning purposes.

For flooding, the researchers used Height Above Nearest Drainage analysis, GloFAS forecast information, and community-mapped flood points for spatial comparison. For drought, they used geospatial climate inputs and a ConvLSTM model. The paper expressly identified Rapid Application Development as its Web GIS

development process; these hydrological, machine-learning, and development procedures served different methodological roles.

The mapped flood points showed qualitative spatial agreement with the flood-model output. The reported median ROC-AUC of 0.87 and median F1 score of 0.81 refer to the drought model, not to the flood analysis. The application brought these multi-hazard outputs together in a Web GIS interface.

The study demonstrates the value of combining spatial analysis, local validation, and accessible map presentation. Its forecast data, modeling choices, and Tanzanian geographic conditions cannot be transferred directly to Bacoor City. FloodSense uses deterministic scenario-based assessment and cannot claim the study's early-warning or machine-learning results as its own.

Near Real-Time Responsive Flood Event Representation: An Open-Source Interactive Web Application Architecture: A Melbourne Digital-Twin Flood Display

Turner and Sun (2024) developed an open-source web-application architecture for representing near-real-time flood events in the Moonee Valley and Maribyrnong areas of Melbourne, Australia. The research problem concerned how river-height and rainfall streams could be joined with a digital city model to display developing events and their potential impacts for decision-makers.

The technical procedure assembled government elevation and building data, administrative boundaries, open water polygons, and Melbourne Water station feeds in a spatial database. A web interface combined these data with event triggers, geospatial queries, and three-dimensional visualization. The source describes this architecture and its proof-of-concept implementation; it does not name an SDLC model.

The authors reported a functioning spatial back end, near-real-time data connections, an interactive front end, and on-demand impact statistics. The paper presents a prototype architecture rather than evidence of resident acceptance, warning accuracy during all floods, or operational deployment throughout Melbourne.

Its combination of spatial storage, data-source identification, and accessible map presentation informs the proposed application's geographic design. The live sensors, impact analytics, Australian data, and emergency-event purpose cannot be transferred to Bacoor City or attributed to the user-requested, hypothetical-scenario assessment (Turner & Sun, 2024).

Related Local Studies

Development of a Water Level Data Logging System with Web-Based Data Visualization for Remote Monitoring Stations: A Water-Level Monitoring and Web-Visualization System

Añana and Beltran (2024) developed a water-level data-logging device with a web-based time-series display to support remote monitoring by local authorities in the Philippines. Their study addressed the presentation of measured water conditions at monitoring stations. It did not assess a resident-selected hypothetical rainfall scenario.

The authors described Systems Development Life Cycle stages of requirements specification, analysis, interface design, development, and testing, without naming a more specific software-development model. They gathered requirements through interviews with barangay and city disaster-risk offices, developed a sensor-linked web chart, and tested water-level display in a tank, at a seawall, and at a riverside. These are documented development and technical procedures, not evidence of a resident-user evaluation.

The reported tank and seawall tests displayed changing water levels and their corresponding categories in the web interface. The riverside test took place under calm conditions and showed a green category; the paper did not evaluate warning performance during a flood. Its documented contribution is remote visualization of sensor readings and category information, not verified reduction of flood harm.

The study illustrates how time-series display and status categories can make measured conditions easier to interpret. Its water-level sensor, test settings, and site-specific category thresholds differ from FloodSense's resident-requested susceptibility assessment and separate preparedness guidance. The tested readings and

thresholds cannot validate classifications for Bacoor City, and the source does not establish system-wide operational effectiveness.

Isabela City Flood Monitoring System Using Arduino and GSM Module: An Arduino and GSM Flood-Monitoring and SMS-Warning System

Carpio, Asnaul, Said, and Amerin (2022) developed an Arduino and GSM flood-monitoring system for Riverside Barangay in Isabela City, Basilan. The system was intended to observe water conditions and communicate warning signals to local disaster personnel. It addressed a local monitoring problem rather than the interpretation of a hypothetical resident-chosen rainfall scenario.

The authors explicitly used an Agile Development Model for incremental system construction and a quantitative survey design for evaluation. Twenty City Disaster Risk Reduction and Management Office personnel participated in a demonstration and initial device test. They assessed functionality, design, security, efficiency, and reliability, making the survey an evaluation of the prototype under the reported conditions.

The paper reported an overall mean of 3.70, interpreted by its authors as Acceptable, and described warning-signal operation during testing. It also reported that battery operation was limited to approximately ten hours. These results describe the tested device and evaluator group; they do not show sustained warning performance across Isabela City.

The Philippine setting provides a useful comparison for local disaster-information systems and for distinguishing development method from evaluation design. However, its hardware, threshold conditions, respondent group, and measured outcomes cannot be transferred to Bacoor City. FloodSense does not continuously monitor water level or send autonomous SMS warnings.

iAlerto: A Web and Mobile Alert System for Pasig City Disaster Risk Reduction Management Office (PCDRRMO) with mobile GPS service integration: A Resident Incident-Reporting and Response Platform

Goh et al. (2023) developed iAlerto for resident incident reporting and PCDRRMO response in Pasig City. The problem was the timely collection of

emergency reports and communication among residents, barangay staff, rescuers, and administrators. Its location function identifies the position of a reported incident rather than computing flood susceptibility from a hypothetical rainfall scenario.

The authors expressly used Scrum for software development and described separate web and mobile roles. They evaluated functionality with 20 information-technology professionals and 20 residents or DRRMO personnel using ISO 9126 functionality criteria. Their source reports the development procedure, users, evaluation method, and system output rather than merely listing programming tools.

The published system included incident reports, GPS-linked location, administrative records, announcements, and evacuation-center management. Respondents gave an overall perceived-functionality mean of 4.48. The authors recommended adding an SMS module for people without internet access, indicating that this feature was not part of the reported evaluated system.

iAlerto is a Philippine example of role-specific emergency information and location-linked reporting. Its Pasig sample, live incident-response function, and perceived functionality result do not establish performance in Bacoor City. FloodSense instead supports pre-event, user-requested susceptibility assessment and separately presented preparedness information (Goh et al., 2023).

Development of an Integrated Natural and Socioeconomic Indicators Monitoring System for Bulacan Using Earth Intelligence Tools: A Web-Based Flood Monitoring and Decision-Support System

Pascual and Leonen (2025) developed an integrated flood-risk monitoring system for Bulacan and evaluated the platform with local government and disaster-risk stakeholders in Meycauayan City. The study sought to combine environmental, geographic, and socioeconomic information in a web-based dashboard for localized risk assessment and decision support. Its system receives observed or externally supplied data rather than a resident-confirmed hypothetical rainfall scenario.

The authors expressly used Design and Development Research as their research design and the Agile Development Lifecycle as their software-development process. They integrated satellite rainfall, dam-discharge information, geographic

hazard layers, and socioeconomic indicators, then evaluated a Random Forest flood-alert classifier using an 80/20 training-and-test split. The paper also reports interface and stakeholder evaluation; these research, development, and technical methods should not be conflated.

The paper reports a web dashboard with geographic flood-risk presentation and a flood-alert classifier with 95% test accuracy, 0.60 precision, 0.73 recall, and a 0.66 F1 score. It also reports stakeholder feedback on navigation and decision-support information. These are results under the authors' Bulacan data and evaluation conditions, not independently established performance in Bacoor City.

The study illustrates how a Philippine local-government information system can combine spatial display, data integration, modeling, and stakeholder review. Its dependence on internet connectivity, gaps in dam telemetry, class imbalance, and location-specific inputs limit transferability. FloodSense does not adopt automatic data ingestion, machine-learning prediction, or autonomous flood alerts, and the Bulacan findings cannot validate its deterministic Bacoor City assessment.

Flood Vulnerability Assessment of Road Network in San Fernando, Pampanga for the Development of Automatic Local Flood Alert and Recording with Mapping Application (ALARMA): A Road-Flood Mapping and Alert Prototype

Geron et al. (2025) assessed road-network flood vulnerability in San Fernando, Pampanga and developed ALARMA to display road conditions and send water-level alerts. The research focused on mobility and flood-related travel disruption. Its road and alert purpose differs from FloodSense's location-specific, pre-event resident susceptibility result.

The authors described an exploratory sequential mixed-methods design, AHP with ratio weighting for rainfall, elevation, waterway proximity, and traffic flow, and QGIS-based spatial analysis. Five government-agency experts provided factor comparisons. The team then built a web map and alert device and used simulated water-level changes and an ISO 25010-based questionnaire to examine the prototype.

The paper reports a mapped road-vulnerability output and website/SMS status changes during 30 simulated update events. No actual flood occurred during that

system test, so the reported simulation performance must not be treated as field accuracy during real floods. The paper also limits its map to four available factors and notes omitted drainage or land-use conditions.

ALARMA demonstrates how local data, expert weighting, geographic display, and prototype checks can be combined. Its Pampanga road network, AHP weights, device thresholds, and alert procedures are not valid substitutes for Bacoor City susceptibility rules. FloodSense neither routes motorists nor issues automatic water-level alerts (Geron et al., 2025).

Comparative Synthesis of Related Studies

Tables 1 and 2 compare five foreign and five Philippine system studies using each source's documented method, objective, and contribution. Table 3 then compares selected system features. The studies differ in geographic coverage, data authority, development procedure, and intended users; their measurements, thresholds, and reported evaluations cannot be transferred directly to Bacoor City.

Table 1. Comparison of related foreign studies

SYSTEM	AUTHOR(S) /YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Design and Implementation of Real-Time Flood Early Warning System (FEWS) Based on IoT Blynk Application: An Arduino- and GSM-Based Mobile Flood-Warning System	Wisudawan (2021)	Site survey; sensor-system design and testing; 30-day reservoir observation of rain, temperature, humidity, and water level.	The study aimed to build and implement a real-time IoT flood early-warning system with mobile access.	It produced four warning states and reported 30-day mean sensor values; no flood occurred during observation.
A User Experience Evaluation of a Mobile Application for Disseminating Site-Specific Impact-Based Flood Warnings: The A4alerts App: A Mobile Application for Site-Specific Impact-Based Flood Warnings	Meléndez-Landaverde and Sempere-Torres (2025)	Online scenario-based user experiment with 76 participants; Likert evaluation, ranking, Friedman test, and Wilcoxon post hoc analysis.	The study evaluated A4alerts' usefulness, suitability, and potential adoption among users.	It found 97% perceived usefulness and ranked site-specific warnings, official warnings, and self-protection guidance highest.

Table 1. Continued

SYSTEM	AUTHOR(S) /YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Real-Time Flood Forecasting and Warning: A Comprehensive Approach toward HCI-Centric Mobile App Development: A Human-Centered Mobile Flood-Forecasting and Warning Prototype	Alsabhan and Dudin (2023)	Activity theory and scenario-based design; nominal groups of five users and five experts; prototype task and satisfaction evaluation.	The study aimed to derive HCI requirements and evaluate a mobile real-time flood-forecasting and warning prototype.	It identified 93 design requirements, three user groups, and mean satisfaction scores from 3.93 to 4.08.
Developing a Climate-Smart Web GIS App for Multi-Hazard Early Warning against Climate-Based Disaster Risks: A Tanzanian Web GIS for Flood and Drought Early Warning	Msusa et al. (2025)	Rapid Application Development (Web GIS); HAND/GloFAS flood analysis and participatory mapping; ConvLSTM drought modeling.	The study aimed to develop a localized multi-hazard Web GIS early-warning application for Tanzania.	It reported strong spatial agreement for flood validation and drought-model median ROC-AUC 0.87 and F1 0.81.
Near Real-Time Responsive Flood Event Representation: An Open-Source Interactive Web Application Architecture: A Melbourne Digital-Twin Flood Display	Turner and Sun (2024)	Technical method: spatial database, live-data API integration, event-triggered geospatial queries, and interactive digital-twin prototyping.	The study aimed to demonstrate an open-source web architecture for near-real-time flood-event display and impact information in Melbourne.	It reported a working spatial back end, sensor-data connections, interactive visualization, and on-demand impact statistics as a proof of concept.

Table 2. Comparison of related local studies

SYSTEM	AUTHOR(S) /YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
Development of a Water Level Data Logging System with Web-Based Data Visualization for Remote Monitoring Stations: A Water-Level Monitoring and Web-Visualization System	Añana and Beltran (2024)	SDLC stages: requirements interviews, analysis, interface design, sensor/web development, and tank, seawall, and riverside tests.	The study aimed to log measured water levels and present time-series trends remotely to support local monitoring.	The web chart displayed test readings and status categories; no flood-event warning performance was evaluated.
Isabela City Flood Monitoring System Using Arduino and GSM Module: An Arduino and GSM Flood-Monitoring and SMS-Warning System	Carpio et al. (2022)	Agile Development Model (software development); quantitative survey and device test with 20 CDRRMO personnel.	The study aimed to develop an Arduino/GSM system that monitors water level and sends SMS warnings.	It obtained an overall mean of 3.70, interpreted as Acceptable, and documented a ten-hour battery limit.

Table 2. Continued

SYSTEM	AUTHOR(S) /YEAR	METHODOLOGY	OBJECTIVES	CONTRIBUTIONS
iAlerto: A Web and Mobile Alert System for Pasig City Disaster Risk Reduction Management Office (PCDRRMO) with mobile GPS service integration: A Resident Incident-Reporting and Response Platform	Goh et al. (2023)	Scrum (software development); functionality assessment by 40 IT, resident, and DRRMO evaluators using ISO 9126 criteria.	The study aimed to improve Pasig residents' incident reporting and PCDRRMO response through linked web and mobile applications.	It reported GPS-linked incident reports, administrative and evacuation-center records, announcements, and a perceived-functionality mean of 4.48.
Development of an Integrated Natural and Socioeconomic Indicators Monitoring System for Bulacan Using Earth Intelligence Tools: A Web-Based Flood Monitoring and Decision-Support System	Pascual and Leonen (2025)	Design and Development Research (research design); Agile Development Lifecycle (software process); spatial integration and Random Forest testing.	The study aimed to integrate environmental and socioeconomic indicators into a Bulacan flood-risk dashboard for local decision support.	It reported a geographic dashboard and flood-alert classifier with 95% test accuracy, 0.60 precision, and 0.73 recall.
Flood Vulnerability Assessment of Road Network in San Fernando, Pampanga for the Development of Automatic Local Flood Alert and Recording with Mapping Application (ALARMA): A Road-Flood Mapping and Alert Prototype	Geron et al. (2025)	Exploratory sequential mixed methods; expert AHP-ratio weighting, QGIS road analysis, web/device prototyping, and simulated alert tests.	The study aimed to map vulnerable San Fernando roads and develop a flood-status and alert application for motorists.	It reported a road-vulnerability map and website/SMS status updates in 30 simulated events; no actual flood occurred during the test.

Table 3 contrasts the documented features of the ten reviewed systems with the approved FloodSense design scope. Most reviewed systems emphasize observed water conditions, forecasts, alerts, or operational incident response; the proposed system instead organizes a user-requested hypothetical rainfall assessment, explained deterministic result, and separate preparedness guidance. The matrix compares purposes and documented functions, not accuracy or implementation maturity. A blank cell means that the inspected publication did not establish the feature.

Table 3. Comparison of developed study to the other existing systems.

FEATURES	**EXISTING SYSTEMS**										**Proposed System**
	S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	**S10**	
User-facing mobile or web interface	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Geographic or map-based presentation		✓		✓	✓			✓	✓	✓	✓
Location-specific information		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
User-selected hypothetical rainfall scenario											✓
Deterministic rule-based susceptibility assessment											✓
Human-readable hazard or assessment context	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Preparedness or self-protection guidance		✓						✓			✓
Administrative management interface								✓			✓

Legend:

✓ - documented in a reviewed system; in the Proposed System column, included in FloodSense's approved design scope, not necessarily completed or evaluated. Blank - not established by the inspected source.

S1 - Design and Implementation of Real-Time Flood Early Warning System (FEWS) Based on IoT Blynk Application

S2 - A User Experience Evaluation of a Mobile Application for Disseminating Site-Specific Impact-Based Flood Warnings: The A4alerts App

S3 - Real-Time Flood Forecasting and Warning: A Comprehensive Approach toward HCI-Centric Mobile App Development

S4 - Developing a Climate-Smart Web GIS App for Multi-Hazard Early Warning against Climate-Based Disaster Risks

S5 - Near Real-Time Responsive Flood Event Representation: An Open-Source Interactive Web Application Architecture

S6 - Development of a Water Level Data Logging System with Web-Based Data Visualization for Remote Monitoring Stations

S7 - Isabela City Flood Monitoring System Using Arduino and GSM Module

S8 - iAlerto: A Web and Mobile Alert System for Pasig City Disaster Risk Reduction Management Office (PCDRRMO) with mobile GPS service integration

S9 - Development of an Integrated Natural and Socioeconomic Indicators Monitoring System for Bulacan Using Earth Intelligence Tools

S10 - Flood Vulnerability Assessment of Road Network in San Fernando, Pampanga for the Development of Automatic Local Flood Alert and Recording with Mapping Application (ALARMA)

METHODOLOGY

This chapter describes the materials, research method, participants, and statistical treatment used to investigate flood-information needs and design and develop FloodSense. Design and Development Research is the study's research design; Evolutionary Prototyping is the software-development model used to construct and refine the application.

Materials

The study used a bilingual resident needs-assessment questionnaire with a consent notice, Google Forms and face-to-face survey materials, exported responses, institutional interview guides, consented recordings and transcripts, request letters, relevant school documents, scholarly and official references, requirements records, interface designs, and software test records. A handwritten task board and Excel tracker supported task tracking; Messenger or Viber, Discord, and email supported coordination and institutional requests.

Development materials included Windows computers, Android test devices, Git and GitHub for sharing and maintaining work, Flutter and Dart for the resident application, Python with Django and Django REST Framework for backend and administration services, PostgreSQL/PostGIS for relational and spatial storage, mapping components, and a browser-based administration interface. These tools supported implementation and technical checks; they did not establish the institutional or scientific validity of the geographic and susceptibility information.

Method



Figure 2. Six-phase DDR framework (Ellis and Levy, 2010).

The study adopted Design and Development Research (DDR). Ellis and Levy (2010) present six research phases: identify the problem, describe the objectives, design and develop the artifact, test the artifact, evaluate testing results, and communicate testing results. Figure 2 is the researchers' illustration of this published sequence for FloodSense. The phases organize the study's inquiry and reporting; they do not imply that planned post-use evaluation or deployment has been completed.

Evolutionary Prototyping is the software-development model within the design-and-development phase. The team constructed and refined the same working prototype through requirements clarification, implementation, internal checks, and relevant review. The handwritten board and Excel tracker recorded tasks; they are not research methods or software-development models (Bjarnason et al., 2023).

Identify the Problem

The approved study concept, literature, and an initial purposive survey of 40 qualified Bacoor City residents established the study's pre-event information problem. The survey used Google Forms and face-to-face collection to record flood experiences, information difficulties, preparedness needs, and technology conditions. It did not ask respondents to evaluate FloodSense after use (see Appendix 4. Initial Resident Needs-Assessment Questionnaire, Blank Copy).

Consented, recorded, and transcribed interviews with BDRRMO and CPDCO supplied institutional context for geographic information, preparedness, source authority, and permission boundaries. The interviews informed problem analysis but did not certify a susceptibility dataset or demonstrate system effectiveness.

Describe the Objectives

The research objectives translated identified resident needs and the approved study scope into requirements for location and hypothetical rainfall input, explainable deterministic susceptibility assessment, separate preparedness guidance, geographic presentation, privacy, and authorized administration. Candidate requirements were checked against source availability, data authority, and technical constraints. The objectives did not establish that every proposed function or information source was operational.

Design and Develop the Artifact

FloodSense is the research artifact. Flutter supports the Android resident interface; Django and Django REST Framework support backend and administrative services; and PostgreSQL with PostGIS supports relational and spatial storage. Through Evolutionary Prototyping, the team continued to refine the same application following internal checks and relevant domain feedback.

The resident assessment flow receives a user-confirmed hypothetical rainfall scenario and location, applies deterministic rule matching and conflict handling, and returns a susceptibility result with a human-readable explanation. Preparedness guidance remains separate from the assessment. Access controls, source provenance, and geographic interpretation govern supporting information; provisional candidate data are not presented as official susceptibility evidence.

Test the Artifact

The team manually checked functions after construction and refinement and used available automated mobile and backend tests for defined behaviors, including access control, input validation, location confirmation, rule matching, results, and supported administrative workflows. These tests establish only specified technical behavior under the tested conditions; they do not establish resident acceptance, scientific validity, or institutional approval.

Evaluate Testing Results

The researchers examined internal test observations against expected behavior, addressed identified discrepancies, and repeated applicable checks on the evolving prototype. This formative review of technical test results is distinct from the planned post-use evaluation of FloodSense by intended users and relevant institutional participants. Product-quality or acceptance conclusions require defined tasks, approved instruments, validated responses, and analysis.

Communicate Testing Results

The manuscript and project records document the initial needs assessment, system design, technical checks, and limitations supported by available evidence. Final post-use findings will be reported only after the evaluation is conducted and analyzed. Neither a working prototype nor a planned evaluation establishes production deployment or an official flood-warning function.

Participants of the Study

Purposive sampling was used for the initial needs assessment. The analyzed dataset contained 40 qualified respondents who consented, were at least 18 years old, lived in Bacoor City, and had experienced flooding or serious heavy-rain effects. They represented 14 of the city's 47 barangays. Niog contributed 13 respondents; Zapote I, II, and III contributed 15; and ten other barangays contributed 12. The concentration of 28 respondents (70.0%) in Niog and the Zapote barangays limits generalization to the whole city. Selected aggregate needs-assessment findings are presented in the survey table (see Appendix Table 2. Initial Needs-Assessment Summary).

Thirty respondents (75.0%) were 18–34 years old. BDRRMO and CPDCO key informants contributed institutional requirements and data-governance context, not resident-user evaluation or evidence of system effectiveness. Participation was governed by the survey consent notice and interview-recording permission. Direct identifiers were excluded from the survey summary, and geographic responses were interpreted with care.

Table 4. Distribution of initial resident-survey respondents

RESPONDENT AREA	FREQUENCY	PERCENTAGE
Niog	13	32.5%
Zapote I, II, and III	15	37.5%
Other ten represented barangays	12	30.0%
Total	40	100.0%

Statistical Treatment of Data

The initial needs-assessment responses were summarized with frequencies and percentages. For multiple-response questions, each option used the 40 qualified respondents as its denominator, so option percentages could exceed 100% when added. The eight ordered information-problem items were also summarized by valid scored-answer means. Never, Rarely, Sometimes, Often, and Always were scored from 1 to 5; “Not sure” was retained in the frequency distribution but excluded from the mean for that item. No sample-size formula, response rate, or post-use system-evaluation score was calculated.

1. Percentage. The proportion selecting a response category or option was computed as follows:

The Formula is:

$$P = \frac{f}{N} \times 100$$

Where:

P = percentage

f = number of respondents selecting the category or option

N = number of qualified respondents (40 for the reported survey items)

2. Item mean. The five-point information-problem responses were summarized as follows:

The Formula is:

$$M = \frac{\Sigma(f_i \times x_i)}{\Sigma f_i}$$

Where:

M = item mean

x_i = score from 1 to 5

f_i = frequency of that score

Σf_i = number of valid scored answers for the item; “Not sure” was excluded from this denominator.

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APPENDICES

Appendix Table 1. Gantt Chart for September-December 2026

PROCESS	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER
Identify the Problem				
Describe the Objectives				
Design and Develop the Artifact				
Test the Artifact				
Evaluate Testing Results				
Communicate Testing Results				
Revisions				

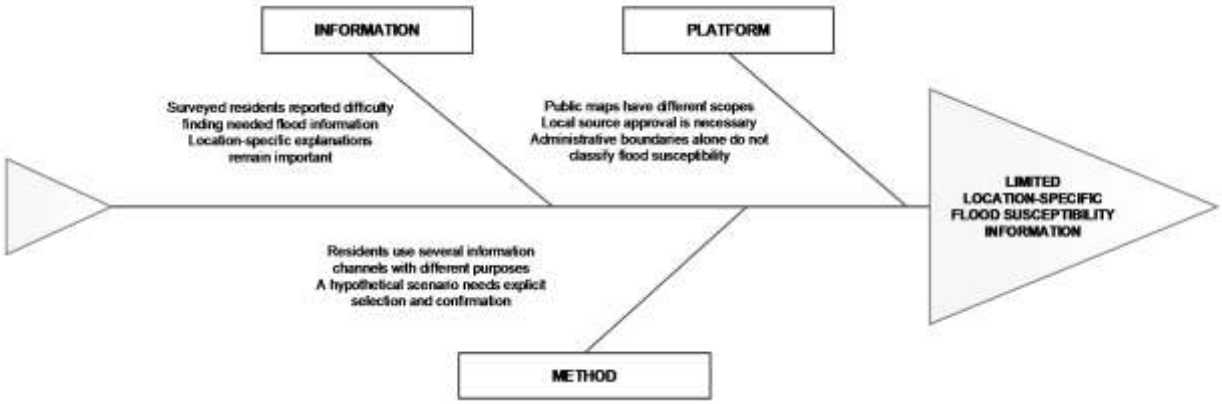
Shading identifies months with documented activity in the listed stage; it does not signify completion of the entire stage. Unshaded months are not recorded as completed activity. Source: dated FloodSense study and development records, September-October 2026.

Appendix Table 2. Initial Needs-Assessment Summary (n=40)

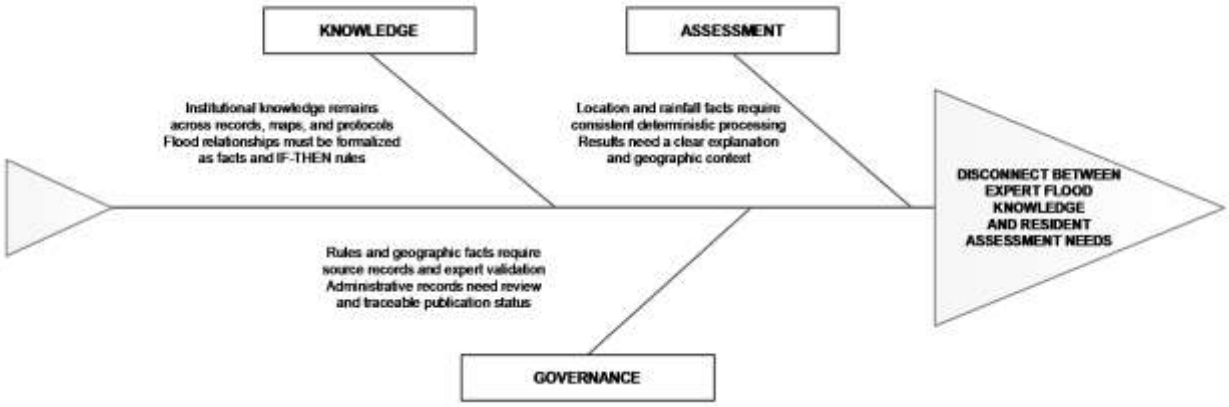
SURVEY INDICATOR	COUNT	PERCENT
Difficulty obtaining needed flood information	29	72.5%
Information inadequacy often or always affected decisions	30	75.0%
Difficulty determining location-specific flood susceptibility	30	75.0%
Difficulty locating verified evacuation-center information	33	82.5%
Regular access to an owned or shared Android device	39	97.5%

Source: FloodSense Initial Survey Results and Summary, October 11, 2026. The 40 respondents were purposively selected from 14 Bacoor City barangays. These descriptive percentages do not estimate prevalence among all Bacoor residents or evaluate FloodSense after use.

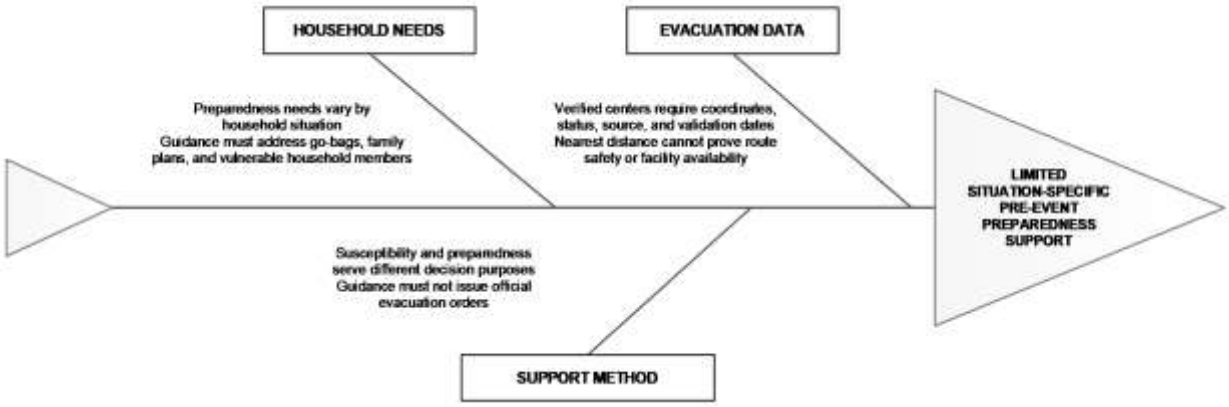
Appendix 1. Fishbone Diagrams



Appendix Figure 1. Fishbone Diagram of Location-Specific Flood Information Needs

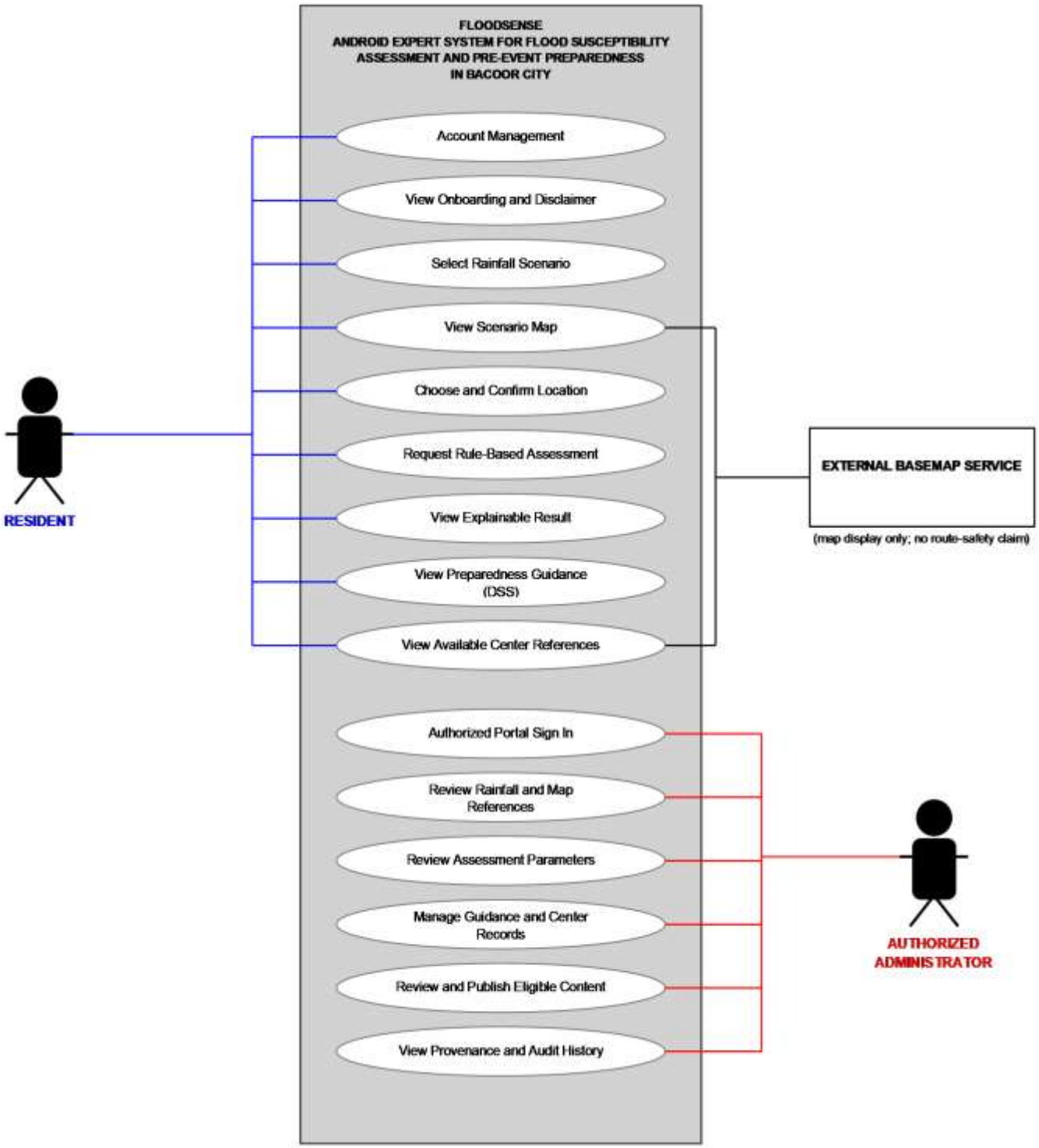


Appendix Figure 2. Fishbone Diagram of Expert-Knowledge and Assessment Needs



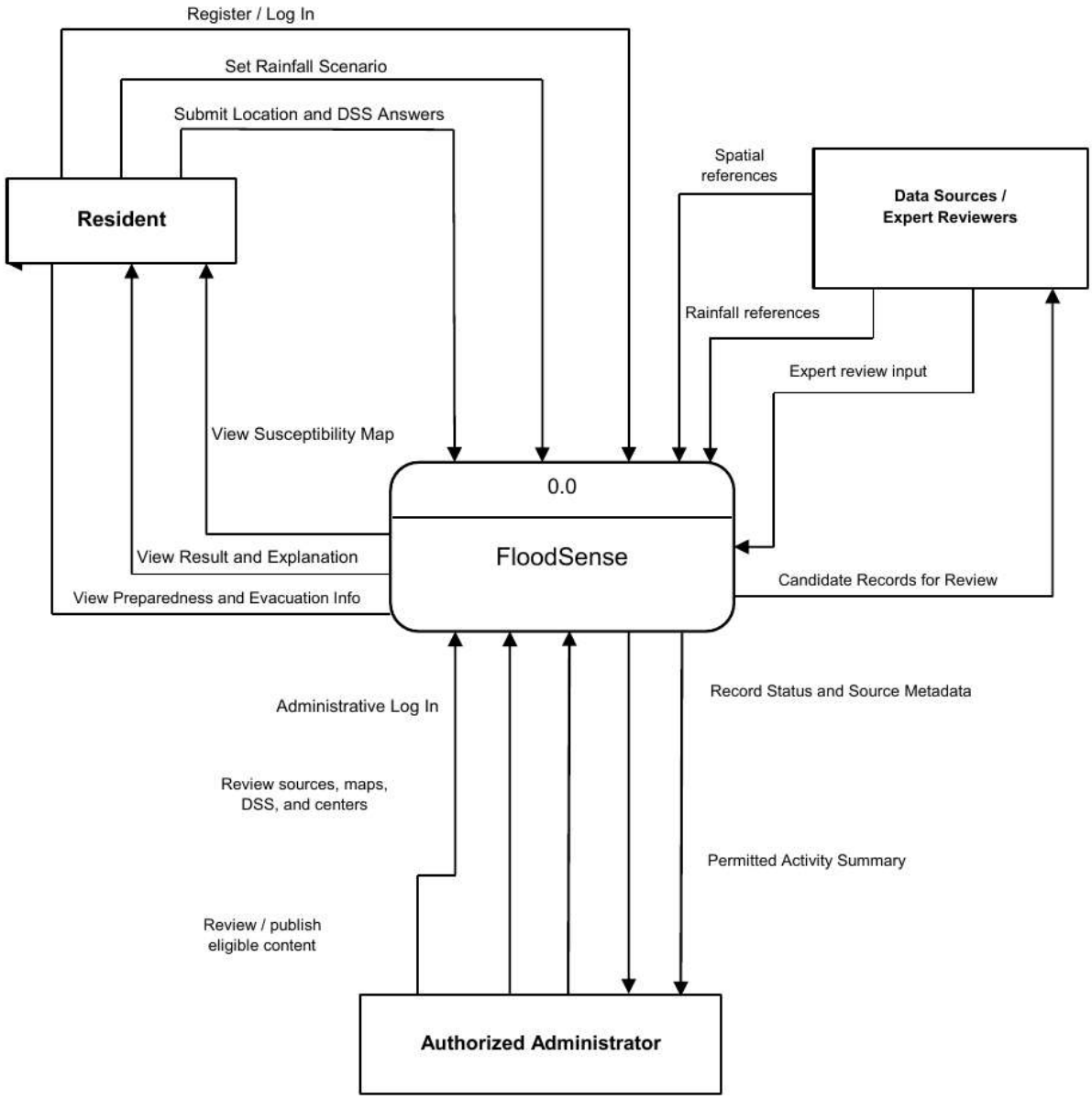
Appendix Figure 3. Fishbone Diagram of Pre-Event Preparedness Needs

Appendix 2. Use Case Diagram



Appendix Figure 4. Use Case Diagram of FloodSense

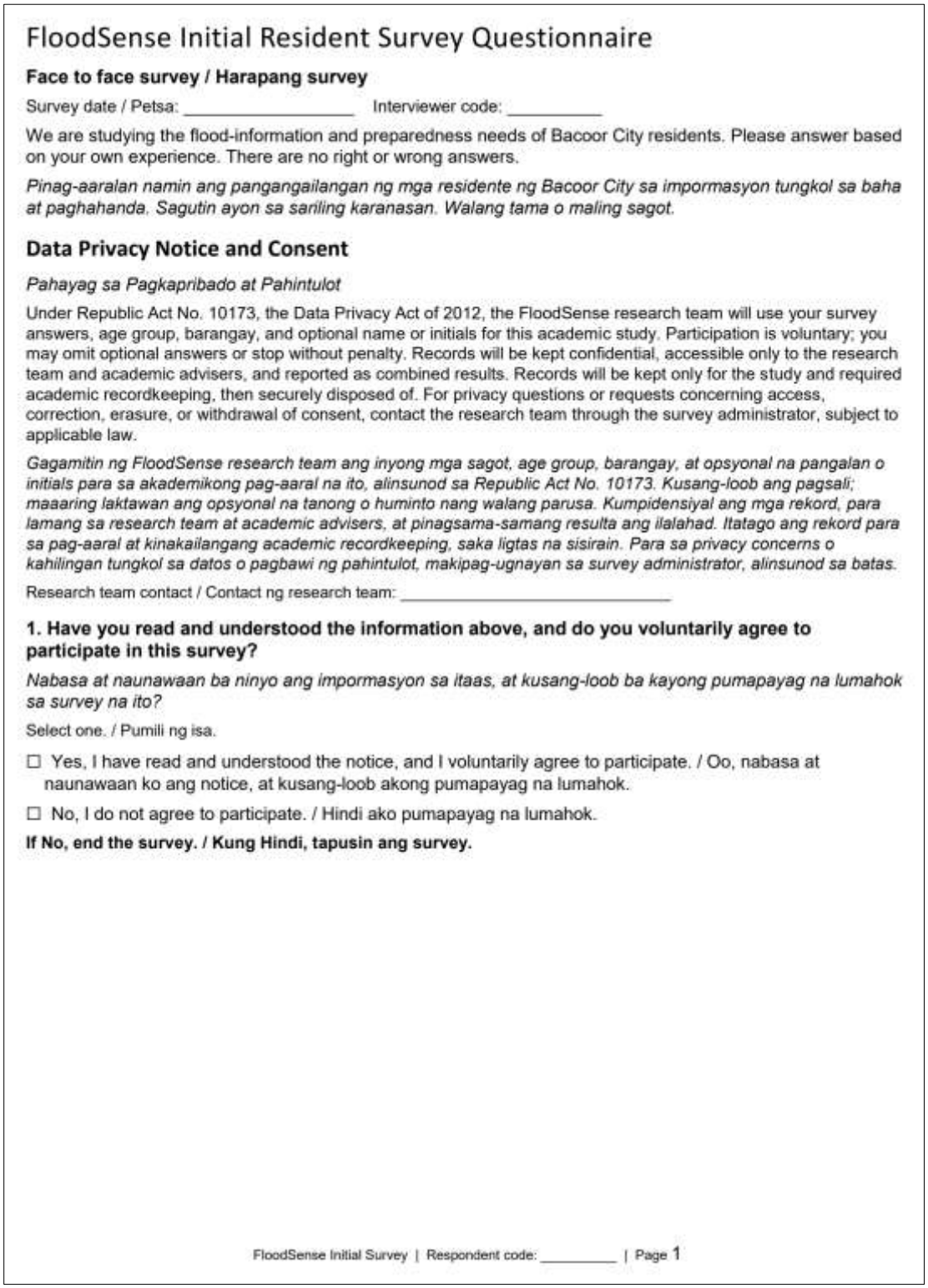
Appendix 3. Context Diagram



Appendix Figure 5. Context Diagram of FloodSense

Appendix 4. Initial Resident Needs-Assessment Questionnaire, Blank Copy

This blank form reproduces the question set represented in the October 11, 2026 response export; it contains no respondent answers.



FloodSense Initial Resident Survey Questionnaire

Face to face survey / Harapang survey

Survey date / Petsa: _____ Interviewer code: _____

We are studying the flood-information and preparedness needs of Bacoor City residents. Please answer based on your own experience. There are no right or wrong answers.

Pinag-aaralan namin ang pangangailangan ng mga residente ng Bacoor City sa impormasyon tungkol sa baha at paghahanda. Sagutin ayon sa sariling karanasan. Walang tama o maling sagot.

Data Privacy Notice and Consent

Pahayag sa Pagkapribado at Pahintulot

Under Republic Act No. 10173, the Data Privacy Act of 2012, the FloodSense research team will use your survey answers, age group, barangay, and optional name or initials for this academic study. Participation is voluntary; you may omit optional answers or stop without penalty. Records will be kept confidential, accessible only to the research team and academic advisers, and reported as combined results. Records will be kept only for the study and required academic recordkeeping, then securely disposed of. For privacy questions or requests concerning access, correction, erasure, or withdrawal of consent, contact the research team through the survey administrator, subject to applicable law.

Gagamitin ng FloodSense research team ang inyong mga sagot, age group, barangay, at opsyonal na pangalan o initials para sa akademikong pag-aaral na ito, alinsunod sa Republic Act No. 10173. Kusang-loob ang pagsali; maaaring laktawan ang opsyonal na tanong o huminto nang walang parusa. Kumpidensiyal ang mga rekord, para lamang sa research team at academic advisers, at pinagsama-samang resulta ang ilalahad. Itatago ang rekord para sa pag-aaral at kinakailangang academic recordkeeping, saka ligtas na sisirain. Para sa privacy concerns o kahilingan tungkol sa datos o pagbawi ng pahintulot, makipag-ugnayan sa survey administrator, alinsunod sa batas.

Research team contact / Contact ng research team: _____

1. Have you read and understood the information above, and do you voluntarily agree to participate in this survey?

Nabasa at naunawaan ba ninyo ang impormasyon sa itaas, at kusang-loob ba kayong pumapayag na lumahok sa survey na ito?

Select one. / Pumili ng isa.

☐ Yes, I have read and understood the notice, and I voluntarily agree to participate. / Oo, nabasa at naunawaan ko ang notice, at kusang-loob akong pumapayag na lumahok.

☐ No, I do not agree to participate. / Hindi ako pumapayag na lumahok.

If No, end the survey. / Kung Hindi, tapusin ang survey.

FloodSense Initial Survey | Respondent code: _____ | Page 1

Appendix Figure 6. Questionnaire Page 1 of 10

Appendix 4. Continued

Eligibility and Respondent Information

Pagiging Kwalipikado at Impormasyon ng Respondent

End the survey if any eligibility answer is No or cannot be confirmed. / Tapusin kung Hindi ang sagot o hindi makumpirma ang pagiging kwalipikado.

2. Are you 18 years old or older?

Kayo ba ay 18 taong gulang o mas matanda?

Select one. / Pumili ng isa.

☐ Yes / Oo ☐ No / Hindi

3. Do you currently reside in Bacoor City?

Kayo ba ay kasalukuyang naninirahan sa Bacoor City?

Select one. / Pumili ng isa.

☐ Yes / Oo ☐ No / Hindi

4. Within the past three years, have you experienced flooding or heavy rainfall that affected your household, community, travel, work, or studies?

Sa loob ng nakaraang tatlong taon, nakaranas ba kayo ng pagbaha o malakas na ulan na nakaapekto sa inyong tahanan, komunidad, pagbiyahe, trabaho, o pag-aaral?

Select one. / Pumili ng isa.

☐ Yes / Oo ☐ No / Hindi

Demographic Information

Impormasyong Demograpiko

5. Name or initials (Optional)

Pangalan o initials (Opsyonal)

Optional. Leave blank if you prefer. / Opsyonal. Maaaring iwang blangko.

_____

6. Which barangay in Bacoor City do you currently reside in?

Saang barangay sa Bacoor City kayo kasalukuyang naninirahan?

Write your barangay only; do not write your exact home address. / Barangay lamang; huwag isulat ang eksaktong address.

Barangay: _____

7. What is your age group?

Ano ang inyong age group?

Select one. / Pumili ng isa.

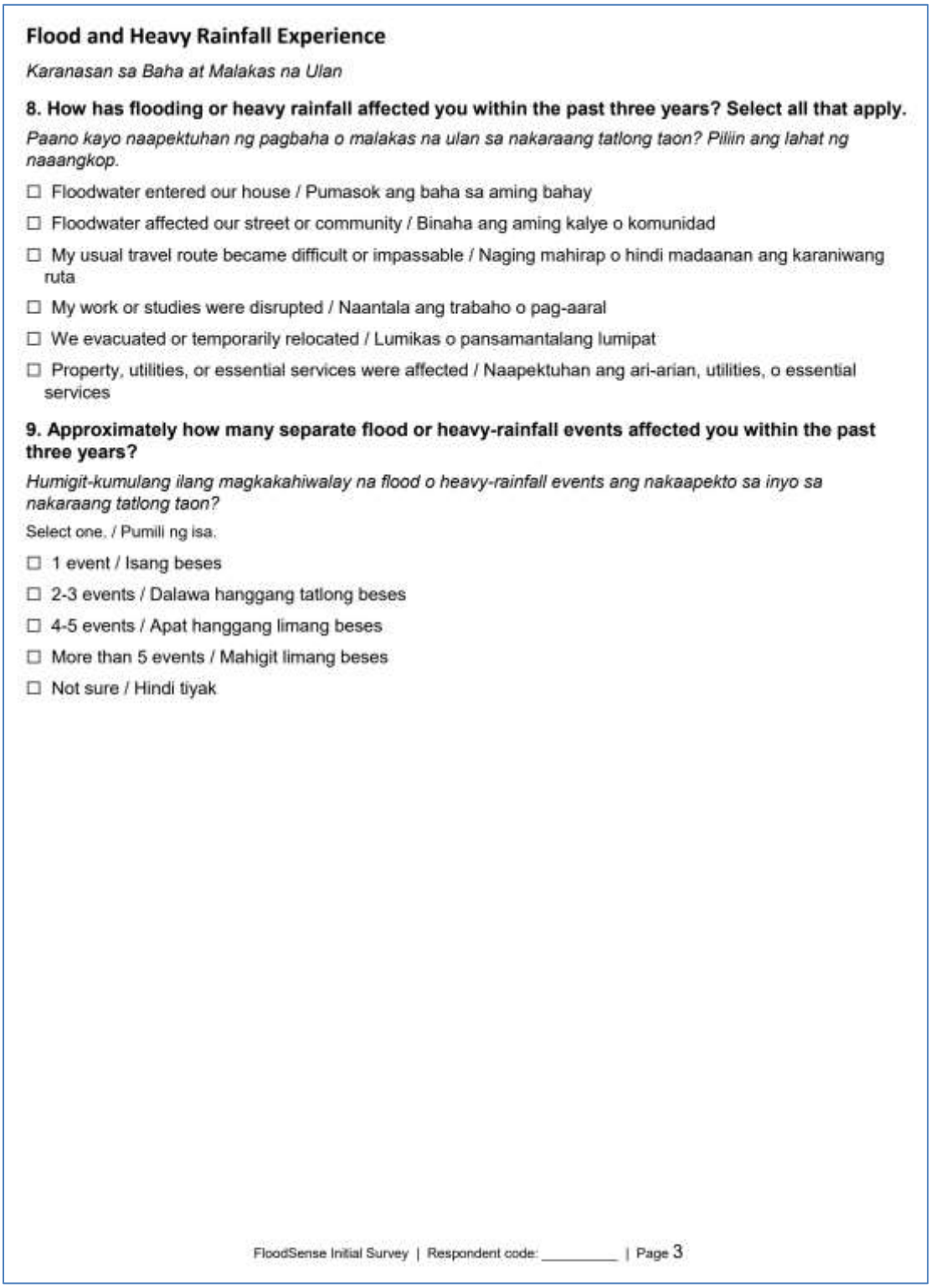
☐ 18-24 years old / taong gulang ☐ 25-34 years old / taong gulang

☐ 35-44 years old / taong gulang ☐ 45-54 years old / taong gulang

☐ 55-64 years old / taong gulang ☐ 65 years old or older / 65 taong gulang o mas matanda

Appendix Figure 7. Questionnaire Page 2 of 10

Appendix 4. Continued



Flood and Heavy Rainfall Experience

Karanasan sa Baha at Malakas na Ulan

8. How has flooding or heavy rainfall affected you within the past three years? Select all that apply.

Paano kayo naapektuhan ng pagbaha o malakas na ulan sa nakaraang tatlong taon? Piliin ang lahat ng naaangkop.

- ☐ Floodwater entered our house / Pumasok ang baha sa aming bahay
- ☐ Floodwater affected our street or community / Binaha ang aming kalye o komunidad
- ☐ My usual travel route became difficult or impassable / Naging mahirap o hindi madaanan ang karaniwang ruta
- ☐ My work or studies were disrupted / Naantala ang trabaho o pag-aaral
- ☐ We evacuated or temporarily relocated / Lumikas o pansamantalang lumipat
- ☐ Property, utilities, or essential services were affected / Naapektuhan ang ari-arian, utilities, o essential services

9. Approximately how many separate flood or heavy-rainfall events affected you within the past three years?

Humigit-kumulang ilang magkakahiwalay na flood o heavy-rainfall events ang nakaapekto sa inyo sa nakaraang tatlong taon?

Select one. / Pumili ng isa.

- ☐ 1 event / Isang beses
- ☐ 2-3 events / Dalawa hanggang tatlong beses
- ☐ 4-5 events / Apat hanggang limang beses
- ☐ More than 5 events / Mahigit limang beses
- ☐ Not sure / Hindi tiyak

FloodSense Initial Survey | Respondent code: _____ | Page 3

Appendix Figure 8. Questionnaire Page 3 of 10

Appendix 4. Continued

Current Flood Information Sources

Kasalukuyang Pinagmumulan ng Impormasyon Tungkol sa Baha

10. Where do you currently obtain flood or heavy-rainfall information? Select all that apply.

Saan kayo kasalukuyang kumukuha ng impormasyon tungkol sa baha o malakas na ulan? Piliin ang lahat ng naaangkop.

If you select a no-problem, none, or not-sure option, select it alone. / Kung wala o hindi tiyak ang sagot, iyon lamang ang piliin.

- ☐ Barangay announcements or public-address system / Anunsyo o public-address system ng barangay
- ☐ Bacoor City or BDRRMO official social-media page
- ☐ PAGASA or another government agency / PAGASA o ibang ahensya ng pamahalaan
- ☐ Television or radio news / Balita sa telebisyon o radyo
- ☐ Online news outlet / Online news
- ☐ SMS or text message / SMS o text message
- ☐ Facebook groups or other social-media posts
- ☐ Family, friends, or neighbors / Pamilya, kaibigan, o kapitbahay
- ☐ Community organization or homeowners' association / Organisasyon ng komunidad o homeowners' association
- ☐ Government website or online bulletin / Website o online bulletin ng pamahalaan
- ☐ I usually receive no flood information / Karaniwan akong walang natatanggap na flood information

11. Overall, how easy or difficult is it for you to obtain the flood information you need?

Sa kabuuan, gaano kadali o kahirap para sa inyo na makuha ang flood information na kailangan ninyo?

Select one. / Pumili ng isa.

- ☐ Very easy / Napakadali
- ☐ Easy / Madali
- ☐ Neither easy nor difficult / Katamtaman
- ☐ Difficult / Mahirap
- ☐ Very difficult / Napakahirap
- ☐ I usually cannot obtain the information / Karaniwan ay hindi ko nakukuha ang impormasyon
- ☐ Not sure / Hindi tiyak

Appendix Figure 9. Questionnaire Page 4 of 10

Appendix 4. Continued

Problems with Existing Information

Mga Problema sa Kasalukuyang Impormasyon

12. Based on your experience, how often do you encounter each problem with existing flood-information sources?

Batay sa inyong karanasan, gaano kadalas ninyo nararanasan ang bawat problema sa kasalukuyang flood-information sources?

Mark ONE box in EACH row. / Lagyan ng tsek ang ISANG kahon sa BAWAT hanay.

1 Never / Hindi kailanman 2 Rarely / Bihira 3 Sometimes / Paminsan-minsan
4 Often / Madalas 5 Always / Palagi NS Not sure / Hindi tiyak

Problem / Problema	1	2	3	4	5	NS
a. Information arrives too late or is already outdated Huli o luma na ang impormasyon	☐	☐	☐	☐	☐	☐
b. Information is difficult to understand Mahirap unawain ang impormasyon	☐	☐	☐	☐	☐	☐
c. Information is too general and not specific to my location Masyadong general at hindi specific sa aking lugar	☐	☐	☐	☐	☐	☐
d. Different sources provide conflicting information Magkakasalungat ang impormasyon mula sa iba't ibang sources	☐	☐	☐	☐	☐	☐
e. The source or date of the information is unclear Hindi malinaw ang source o petsa ng impormasyon	☐	☐	☐	☐	☐	☐
f. Information is difficult to access when internet or electricity is unreliable Mahirap ma-access kapag mahina ang internet o kuryente	☐	☐	☐	☐	☐	☐
g. Preparedness instructions are too general for my situation Masyadong general ang preparedness instructions	☐	☐	☐	☐	☐	☐
h. Verified evacuation-center information is difficult to find Mahirap mahanap ang verified evacuation-center information	☐	☐	☐	☐	☐	☐

Appendix Figure 10. Questionnaire Page 5 of 10

Appendix 4. Continued

Effects on Decisions and Local Information

Epekto sa Desisyon at Lokal na Impormasyon

13. How often have late, unclear, incomplete, or conflicting flood information affected your decisions?

Gaano kadalas nakaapekto sa inyong mga desisyon ang late, unclear, incomplete, o conflicting flood information?

Select one. / Pumili ng isa.

- ☐ Never / Hindi kailanman
- ☐ Rarely / Bihira
- ☐ Sometimes / Paminsan-minsan
- ☐ Often / Madalas
- ☐ Always / Palagi
- ☐ Not sure / Hindi tiyak

14. Which decisions have become difficult, uncertain, or delayed because of insufficient flood information? Select all that apply.

Aling mga desisyon ang naging mahirap, hindi tiyak, o naantala dahil sa kulang na flood information? Piliin ang lahat ng naaangkop.

If you select a no-problem, none, or not-sure option, select it alone. / Kung wala o hindi tiyak ang sagot, iyon lamang ang piliin.

- ☐ Preparing emergency supplies / Paghahanda ng emergency supplies
- ☐ Moving important belongings to a safer place / Paglilipat ng mahahalagang gamit
- ☐ Deciding whether it is safe to travel / Pagpapasya kung ligtas bumiyahe
- ☐ Choosing a road or area to avoid / Pagpili ng daan o lugar na iiwasan
- ☐ Deciding whether or when to evacuate / Pagpapasya kung lilikas o kailan lilikas
- ☐ Identifying an evacuation center / Pagtukoy ng evacuation center
- ☐ Deciding whether to attend work or school / Pagpapasya tungkol sa trabaho o pag-aaral
- ☐ Helping children, older persons, persons with disabilities, or other household members / Pagtulong sa mga kasambahay na nangangailangan ng tulong
- ☐ None of the above / Wala sa mga nabanggit

Appendix Figure 11. Questionnaire Page 6 of 10

Appendix 4. Continued

Location Specific Information and Priorities

Impormasyon para sa Lugar at mga Prayoridad

15. How difficult is it for you to determine whether your specific barangay or location may be susceptible to flooding before or during heavy rainfall?

Gaano kahirap para sa inyo na malaman kung maaaring flood-susceptible ang inyong partikular na barangay o lokasyon bago o habang may malakas na ulan?

Select one. / Pumili ng isa.

- ☐ Very easy / Napakadali
- ☐ Easy / Madali
- ☐ Neither easy nor difficult / Katamtaman
- ☐ Difficult / Mahirap
- ☐ Very difficult / Napakahirap
- ☐ I do not know how to determine it / Hindi ko alam kung paano ito matutukoy
- ☐ Not sure / Hindi tiyak

16. How easy or difficult is it for you to find verified evacuation-center information relevant to your location?

Gaano kadali o kahirap para sa inyo na makahanap ng verified evacuation-center information na relevant sa inyong lokasyon?

Select one. / Pumili ng isa.

- ☐ Very easy / Napakadali
- ☐ Easy / Madali
- ☐ Neither easy nor difficult / Katamtaman
- ☐ Difficult / Mahirap
- ☐ Very difficult / Napakahirap
- ☐ I do not know where to find it / Hindi ko alam kung saan ito makikita
- ☐ I have not needed this information / Hindi ko pa kinailangan ang impormasyong ito
- ☐ Not sure / Hindi tiyak

17. Which types of information are most important to you before possible flooding? Select up to three.

Aling mga uri ng impormasyon ang pinakamahalaga sa inyo bago ang posibleng pagbaha? Pumili ng hanggang tatlo.

- ☐ Flood susceptibility of my barangay or selected location / Flood susceptibility ng aking barangay o selected location
- ☐ Possible effects of rainfall intensity and duration / Posibleng epekto ng lakas at tagal ng ulan
- ☐ Current road or travel conditions / Kasalukuyang kondisyon ng daan o biyahe
- ☐ Pre-event household preparedness actions / Mga paghahanda ng household bago ang baha
- ☐ Evacuation-center references / Evacuation-center information
- ☐ Map-based visualization of the selected location and susceptibility result / Map visualization ng napiling lokasyon at susceptibility result
- ☐ Verified evacuation-center names and map locations / Verified evacuation-center names at map locations

Appendix Figure 12. Questionnaire Page 7 of 10

Appendix 4. Continued

Preparedness Guidance and Evacuation Details

Gabay sa Paghahanda at Detalye ng Evacuation Center

18. Based on the flood-preparedness advice you currently receive from barangay/LGU announcements, government agencies, news, or social media, what problems have you experienced? Select all that apply.

Batay sa flood-preparedness advice na kasalukuyan ninyong natatanggap mula sa barangay/LGU, government agencies, balita, o social media, anong mga problema ang inyong naranasan? Piliin ang lahat ng naaangkop.

If you select a no-problem, none, or not-sure option, select it alone. / Kung wala o hindi tiyak ang sagot, iyon lamang ang piliin.

- ☐ The guidance is too general / Masyadong general ang guidance
- ☐ It is not specific to my location or situation / Hindi specific sa aking lugar o sitwasyon
- ☐ The recommended actions are difficult to understand / Mahirap unawain ang recommended actions
- ☐ It is unclear which action should be done first / Hindi malinaw kung ano ang unang gagawin
- ☐ The source of the guidance is unclear / Hindi malinaw ang source
- ☐ The guidance is difficult to access / Mahirap itong ma-access
- ☐ I do not usually receive preparedness guidance / Karaniwan akong walang natatanggap na guidance
- ☐ I have not experienced a problem with available guidance / Wala akong naranasang problema
- ☐ Not sure / Hindi tiyak

19. Which evacuation-center details would be useful to you? Select all that apply.

Aling evacuation-center details ang magiging kapaki-pakinabang sa inyo? Piliin ang lahat ng naaangkop.

If you select a no-problem, none, or not-sure option, select it alone. / Kung wala o hindi tiyak ang sagot, iyon lamang ang piliin.

- ☐ Name of the center / Pangalan ng center
- ☐ Address or map location / Address o map location
- ☐ Approximate straight-line distance from a selected location / Approximate straight-line distance
- ☐ Contact information / Impormasyon sa pakikipag-ugnayan
- ☐ Information source or responsible office / Source o responsible office
- ☐ Date the information was last verified / Petsa ng huling verification
- ☐ I do not need evacuation-center information / Hindi ko kailangan ang impormasyong ito
- ☐ Not sure / Hindi tiyak

Appendix Figure 13. Questionnaire Page 8 of 10

Appendix 4. Continued

Technology Access and Preferences

Access sa Teknolohiya at mga Kagustuhan

20. Do you have regular access to an Android smartphone?

May regular access ba kayo sa Android smartphone?

Select one. / Pumili ng isa.

- ☐ Yes, my own device / Oo, sarili kong device
- ☐ Yes, a shared device / Oo, shared device
- ☐ No / Hindi
- ☐ Not sure whether it is Android / Hindi tiyak kung Android

21. How reliable is your internet access during heavy rainfall?

Gaano ka-reliable ang inyong internet access kapag may malakas na ulan?

Select one. / Pumili ng isa.

- ☐ Always reliable / Palaging reliable
- ☐ Usually reliable / Kadalasang reliable
- ☐ Sometimes reliable / Paminsan-minsang reliable
- ☐ Rarely reliable / Bihirang reliable
- ☐ No internet access / Walang internet access
- ☐ Not sure / Hindi tiyak

22. Which location-selection methods would you be willing to use in an application? Select all that apply.

Aling location-selection methods ang handa ninyong gamitin sa isang application? Piliin ang lahat ng naaangkop.

If you select a no-problem, none, or not-sure option, select it alone. / Kung wala o hindi tiyak ang sagot, iyon lamang ang piliin.

- ☐ Manual barangay selection / Manual na pagpili ng barangay
- ☐ Temporary map pin that is not saved as location history / Temporary map pin na hindi ise-save bilang location history
- ☐ One-time GPS access after an explanation and permission request / One-time GPS access matapos ang paliwanag at permission request
- ☐ I am not comfortable providing or selecting a location / Hindi ako komportableng magbigay o pumili ng lokasyon
- ☐ Not sure / Hindi tiyak

The proposed system does not require continuous GPS tracking or background location monitoring. / Hindi nangangailangan ang proposed system ng tuloy-tuloy na GPS tracking o background location monitoring.

23. Which language presentation would be easiest for you to understand?

Aling language presentation ang pinakamadali ninyong mauunawaan?

Select one. / Pumili ng isa.

- ☐ English
- ☐ Filipino
- ☐ Both English and Filipino / Parehong English at Filipino

Appendix Figure 14. Questionnaire Page 9 of 10

Appendix 4. Continued

Additional Information Needs

Karagdagang Pangangailangan sa Impormasyon

24. What is the most important flood information or application function that you need but currently find difficult to obtain?

Ano ang pinakamahalagang flood information o application function na kailangan ninyo ngunit mahirap makuha sa kasalukuyan?

Optional. Write your answer, or tell it to the interviewer. / Opsyonal. Isulat ang sagot o sabihin sa tagapanayam.

_____

_____

_____

_____

_____

_____

_____

_____

Thank you for sharing your experience. / Salamat sa pagbabahagi ng inyong karanasan.

Interviewer: Check for one answer in each single-choice question and each matrix row, and no more than three answers in Question 17. Clarify conflicting selections neutrally; do not suggest an answer. Leave unanswered items blank.

Tagapanayam: Suriin ang tig-isang sagot sa single-choice questions at bawat matrix row, at hanggang tatlong sagot sa Tanong 17. Linawin ang magkasalungat na sagot nang walang pagkiling. Iwang blangko ang hindi sinagutan.

Privacy notice legal reference: Republic Act No. 10173, National Privacy Commission, https://privacy.gov.ph/data-privacy-act-/

Appendix Figure 15. Questionnaire Page 10 of 10